灼智® EBC-A40L
Battery Tester



EBC-A40L User Guide

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⚠ Safety:

- Ensure the machine is grounded to avoid electrical hazards.
- Do not operate the machine in a flammable environment, or in environments with high temperatures or humidity.
- Use the machine only with the specified battery types and voltages, never test abnormal batteries.
- Do not modify the machine or its components without authorization.

Precautions:

Before using this tester, please pay attention to the following:

- Carefully read the user manual before testing.
- Place the tester on a level and sturdy surface.
- Ensure that the ventilation openings of the tester are unobstructed and regularly clean the dust from them.
- **Power on the device before connecting it with a battery and ensure the battery is disconnected before powering off the device. Ensure stable power supply during the testing process.**
- Use the original power cord, connection cable, and test clip cable provided by the manufacturer.
- Do not reverse the polarity of the test clip cables.
- The power socket for the device must be grounded, to avoid bringing sources of static electricity to the metal casing.
- Supervise during testing, and keep the battery separate and insulated

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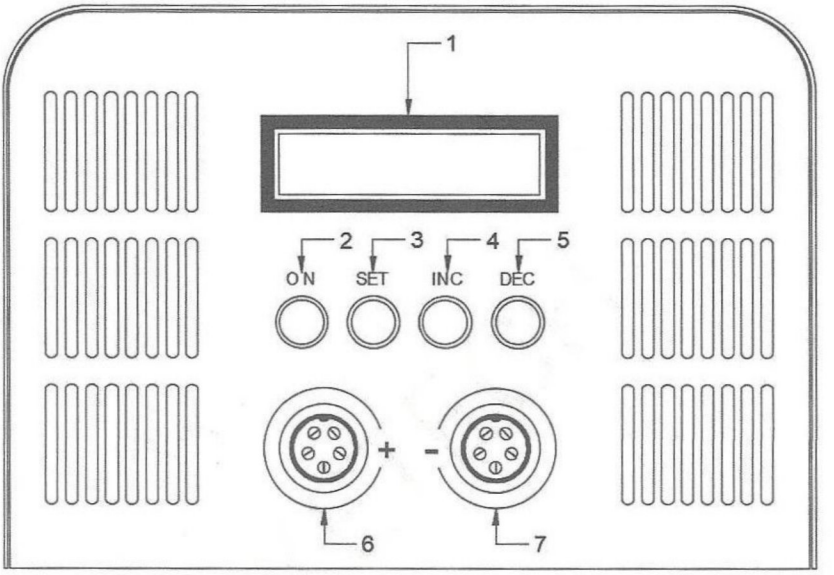
from surrounding objects.

- Whenever abnormal voltage, current data, or other abnormal condition is detected during a test, immediately stop the device manually, check the wiring and battery, and ask for after-sales support.
- Fill out the information on the warranty card before sending it back with the device for after-sales service.

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1. Introduction

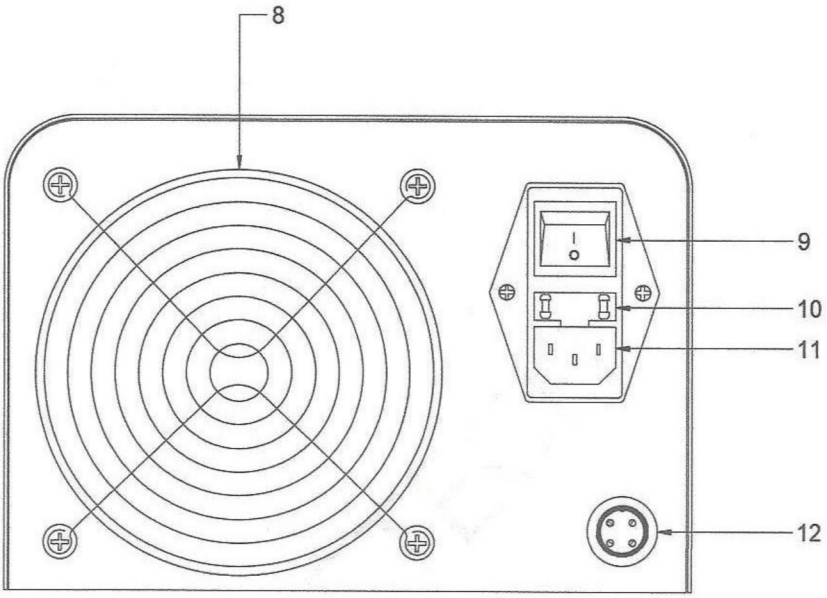
1.1 Front panel



1. The LCD Display Screen
2. Start and stop button
3. Set and switch button
4. Increase button
5. Decrease button
6. Battery positive input
7. Battery negative input

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1.2 Back panel



8. Fan

9. Power switch

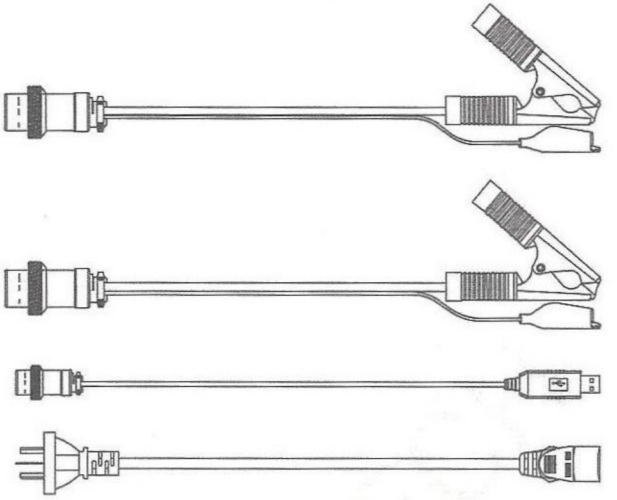
10.Fuse

11.Power cable socket

12.PC cable socket

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1.3 Accessories



Black test cable*1, red test cable*1

Four-core connection cable*1

Power cable*1

2. Features

Thank you for selecting the EBC-A40L. This device is designed for charging and discharging various types of batteries within a 5V range, including lithium battery, lithium iron, 2V / 4V lead acid and other batteries. Current and voltage can be freely set within the range. The device supports single charge, single discharge and charge-discharge-charge cycling test. It also supports computer connection for curve plotting and more than one cycling tests.

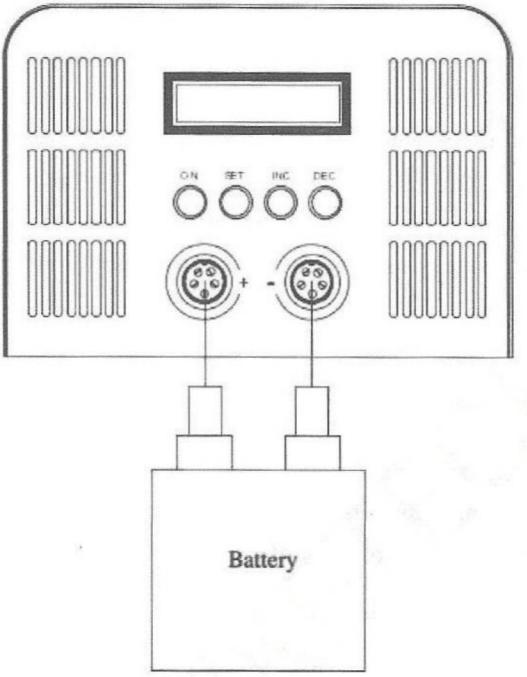
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3. Specifications

Voltage range	0-5V
Voltage accuracy	0.2%±0.03V
Current range	0.1-40A
Current accuracy	0.2%±0.01A
Discharge mode	Constant-current / constant-power discharge
Discharge terminate	Time/ voltage below the set limit / manual
Discharge power	Max 200W
Charging mode	CV constant current and constant voltage charging
Charge terminate	Current below the set limit / manual
Capacity accuracy	Within 10Ah, resolution of 0.001Ah 10Ah – 100Ah, resolution of 0.01Ah Above 100Ah, resolution of 0.1Ah
Heat dissipation	Fan active cooling
Power supply	AC 200-240V (110V can be customized)

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4. Testing connection



4.1 Battery connection

Two battery connectors on the front panel, consist of two 5-pin aviation sockets, with the left positive and the right negative. Batteries can be connected with the device through two larger clips and two smaller clips. Two red clips need to be connected with the battery positive terminal, and two black clips with the negative terminal.

Note: Connect all clips directly with the battery terminals. Avoid clamping the smaller crocodile clip over the larger clip, which will affect the accuracy.

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4.2 Computer connection

This device can be connected with a computer through a USB-TTL cable. After plug the 4-pin aviation cable into the socket on the back panel and successfully connect the device with ZKETECH's EB software, parameters setting and on/off switches can be controlled through software, data and curves will be recorded as well. Before connecting the device with the software, please make sure that the device is in a stopped state (LCD displays OFF). If the test has started by buttons, connection with software will be disabled.

⚠ Warning

Check the battery polarity and voltage carefully before clamping the batteries. Incorrect wiring can result in inaccurate measurements, which can subsequently damage the device or the battery.

4.3 Power supply

Power cable can be connected with a 3-pin AC socket on the back panel.

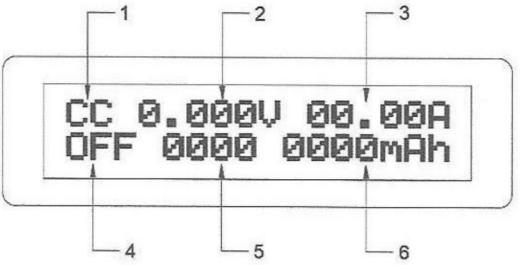
5. Display and setting

The device has two types of interfaces, testing interface and setting interface. Upon startup, it automatically enters the testing interface, which displays voltage, current and other testing data. From the testing interface, press and hold the "SET" button for over two seconds to access the settings interface. In the setting interface, parameters can be modified using the four buttons. After complete setting, return to the testing interface with a long-press on the "SET" knob, and start testing by

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pressing "ON" button.

5.1 Testing interface



1.Test mode

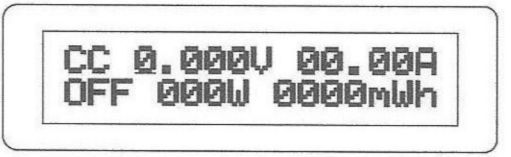
2.Measured voltage

3.Measured current

4.Test status (ON/OFF, DSC or CHG for discharge and charge. AT1 means the first step during charge-discharge-charge cycling test)

5.Test time duration (minutes)

6.Measured capacity (0000 mAh, 00.00Ah and 000.0Ah, switch automatically)



During the test, press the "SET" button can switch display between capacity (mAh) and energy (mWh).

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Press the "ON" button in the test interface to start testing, and press it again to stop.

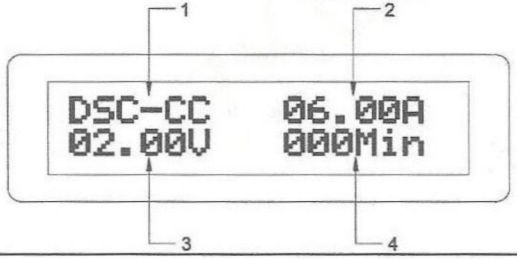
5.2 Setting interface

In the testing interface (OFF state), a long-press on the "SET" button for over 2 seconds will move to the setting interface.

Note: The buttons' function will be disabled once the device is connected to the EB software.

The cursor blinks at the mode. You can press the "INC" or "DEC" button to modify the mode. A short-press on the "SET" button, will move the cursor one space toward right and one press on the "ON" button will move one larger step. Press the "INC" or "DEC" button to modify parameters. If the set parameter exceeds device range, it will automatically be restricted.

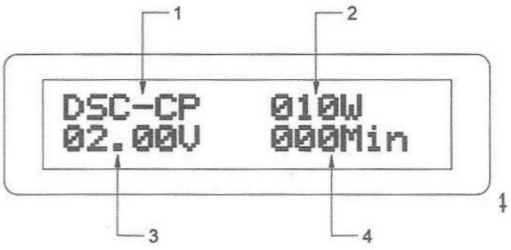
5.2.1 Constant-current discharge



1. Mode (Constant current discharge); 2. Discharge current

3. Termination voltage; 4. Test duration

5.2.2 Constant power discharge



Constant power discharge

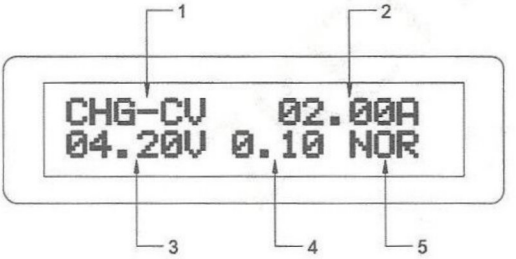
1. Mode (Constant power discharge); 2. Discharge power
3. Termination voltage; 4. Test duration

Note: 1.Test will automatically stop when the measured voltage drops below the termination voltage.

2.Test duration can be set up to 999 minutes.

3.Test can only run within device range, and parameters set beyond allowed range will be restricted automatically.

5.2.3 Constant-current&voltage charge



1. Mode (Charging); 2. Charge current
3. Constant voltage/ termination voltage 4. termination current
5. Cycle state (NOR means single test, AUTO means cycle test)

✧ NOR —— single mode, test stops at termination parameter

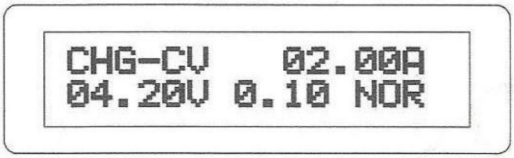
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✧ AUTO —— automatic cycle mode, a three-step cycle test contains "charge-discharge-charge"

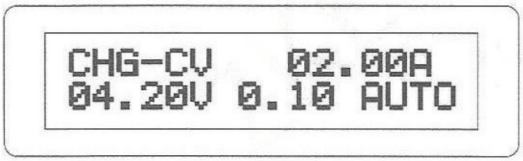
Note: Charge test contains two parts, first constant-current charge till reaching the termination voltage. Second constant-voltage part, the current starts dropping and charge test will stop when measured current drops below the termination current.

5.2.4 Automatic "charge-discharge-charge" test

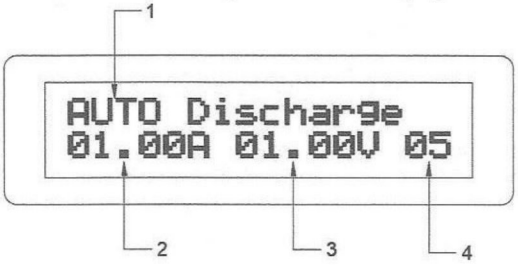
(1) Press the "SET" button for over 2 seconds into the setting interface.



(2) Set charge current, termination voltage and current. Move the cursor to NOR and press the "INC" or "DEC" button to modify to AUTO mode.



(3) When the cursor blinks at AUTO, press the "ON" button for over 2 seconds into next step to set discharge parameters.

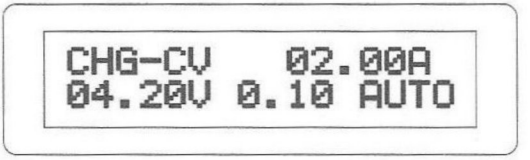


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1.AUTO (means cycle); 2. Discharge current;

3. Termination voltage;4.Interval time (minutes) between two tests

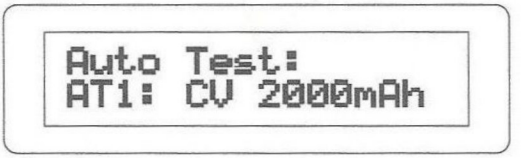
(4) After complete setting discharge parameters, press the "SET" button for over 2 seconds back to the charge setting interface.

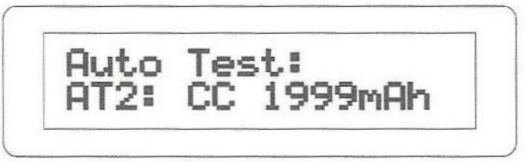


(5) Keep pressing the "SET" button for over 2 seconds and return to the testing interface. Press the "ON" button to start testing, and press it again to stop.

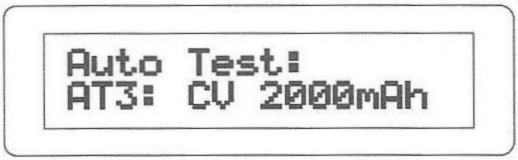
Note: Interval time is recommended to set between two steps, often between 5 and 10 minutes, to help battery cooling.

(6) When the test ends, press "INC" or "DEC" button to read test results of each step:





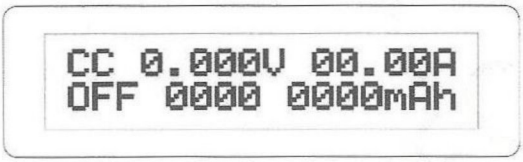
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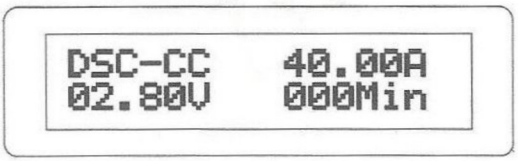
6. Setting examples

6.1 Discharge test (discharge 3.7V lithium battery to 2.8V at 40A):

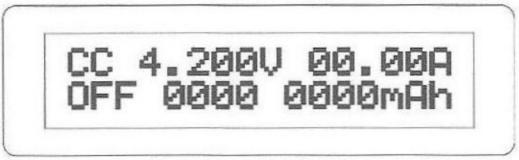
(1) Plug in the power cord, turn on the device, and you can see interface displays as below.



(2) Press the "SET" button for over 2 seconds to enter the setting interface. Modify and set parameters.

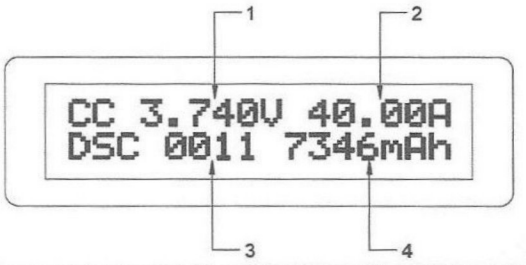


(3) Long press the "SET" button and return to the testing interface. Connect the battery with the device, which shows measured voltage.



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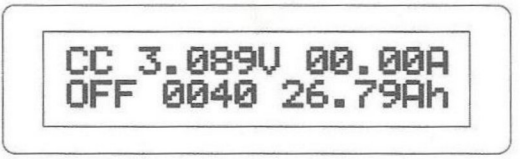
(4) Press the "ON" button shortly to start testing, below displays results after a 11-munite test:



1. Battery voltage; 2. discharge current

3. Discharged Time; 4. Discharged capacity

(5) When the measured voltage reaches the termination voltage, the test stops, the display display OFF-state and the capacity.

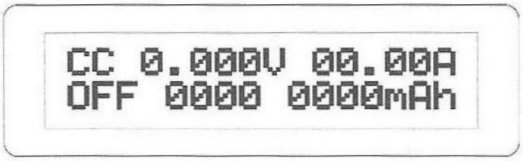


Note: After discharge test stops, the voltage of a healthy battery gradually recovers and rises. Differences in the affinity of metals in the electrodes produce this voltage potential even when the battery is empty. (For example lithium battery will generally rise to 3.3V, etc.).

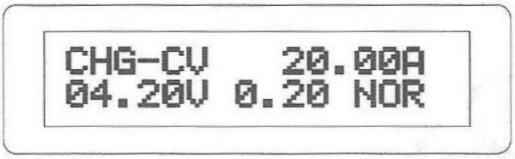
6.2 Charge test (charge 3.7V lithium battery to 4.2V at 20A, termination current at 0.2A):

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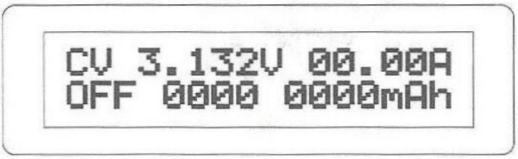
(1) Plug in the power cord, turn on device, and the test interface appears.



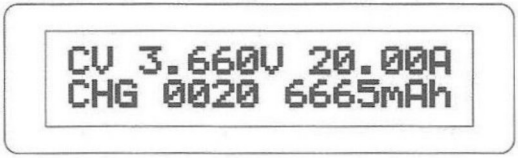
(2) Press the "SET" button and hold on for over 2 seconds to enter the setting interface, move cursor to certain place by pressing "SET" or "ON" button and press "INC" or "DEC" to modify parameters:



(3) When complete parameter setting, long press the "SET" button back to the testing interface. Connect the battery to the device through clips.

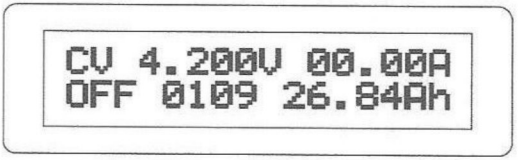


(4) Short press the "ON" button to start testing, below displays a 20-minute test:



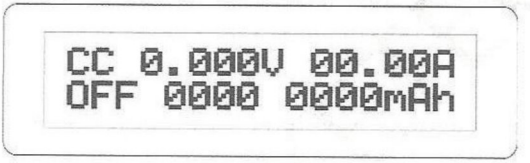
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(5) During the test, when the voltage reaches termination voltage/constant voltage, the test current starts dropping and test won't stop until the measured current drops to termination current.

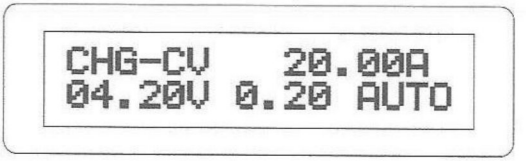


6.3 Charge-discharge-charge test example (charge 3.7V lithium battery to 4.2V at 20A, termination current at 0.2A and discharge to 2.8V at 20A, with a 5-minute interval)

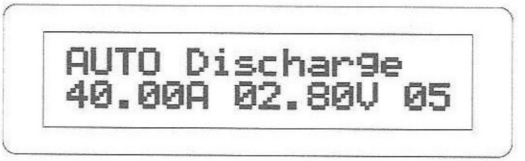
(1) Plug in the power cord, turn on the device, testing interface appears.



(2) Long press the "SET" button for 2 seconds to enter the setting interface. First set charge mode.

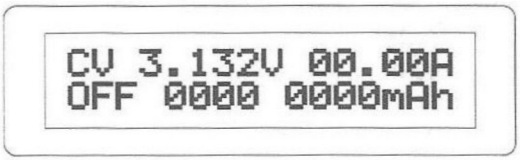


(3) Move the cursor to the AUTO, long press the "ON" button, to enter discharge setting. Set the discharge parameters as below:

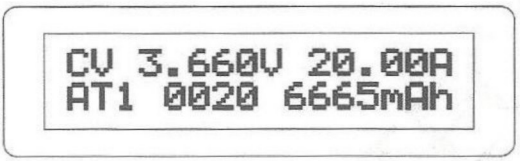


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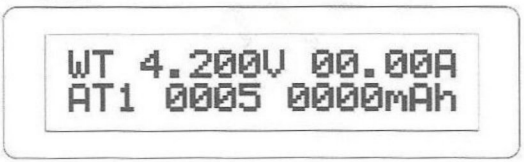
(4) Long press the "SET" button, return to the setting interface, long press the "SET" button again, and return to the testing interface. Connect the battery to the device and it displays battery voltage on the screen:



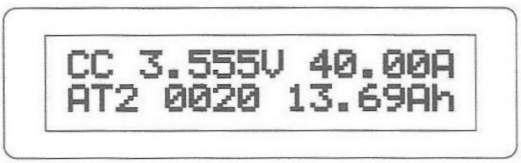
(5) Short press the "ON" button to start testing, AT1 refers to the first step in the cycle test of charge-discharge-charge.



(6) After charge completes, the device will be suspended for five minutes.

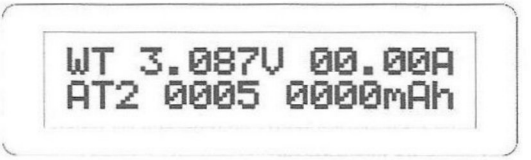


(7) Then AT2 test of discharge starts automatically.

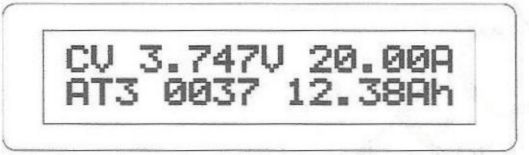


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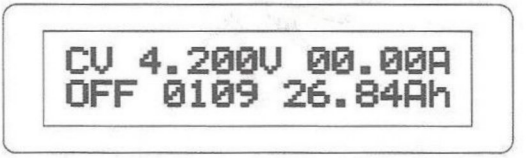
(8) After the discharge test completes, the device will be suspended for five minutes once again.



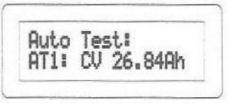
(9) Then AT3 test of charge starts automatically.

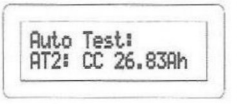


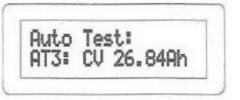
(10) When charge current reaches the termination current, the device stops testing automatically.



(11) After the test stops, press "INC" or "DEC" button to read the test result of each step:







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FAQ

1. After connecting the battery to the device, there is no voltage displayed on the screen?

Check whether the small red and black crocodile clips are connected to the positive and negative terminals of the battery.

2. Start testing but no current displayed on the tester?

Ensure that the large red and black crocodile clips are securely connected to the positive and negative terminals of the battery.

3. The tester cannot start discharge process?

Check whether the termination voltage is set correctly. When the detected voltage is lower than the termination voltage, the tester won't start discharging test.

4. The tester can't connect with a computer?

A) Install the software and driver correctly.

B) Select the corresponding port on the software (can check the port number in the computer's Device Manager, the device can only be connected with the software when the port is between COM2 and COM9)

C) The tester should be in the stop state (if test has been initiated using the buttons, software function is disabled).

5. How to test the capacity?

First charge the battery to full, and then switch to discharge the battery.

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Capacity will display on the screen when the test stops.

6. How to do the cycle testing?

When you control testing through the EB software, you can set up more than one cycle. But if you start testing using buttons, you can set just a charge-discharge-charge cycle.

7. What parameter should be set?

The battery charge and discharge parameters should be set according to the battery specification sheet. Incorrect parameters will damage battery and tester and cause safety concerns. The following are common parameters (follow battery specification sheet is highly recommeded):

3.7V Lithium battery: charging current ≤ 0.5C, charging voltage=4.2V, discharge current≤0.5C, discharge voltage=2.8V.

3.2V lithium iron: charging current ≤ 1C, charging voltage=3.65V, discharge current≤1C, discharge voltage=2.5V.

2V lead acid: charging current≤0.25C, charging voltage=2.4V, discharge current≤0.5C, discharge voltage=1.75V.

*C refers to C-rate of a battery. For a 10Ah battery, 0.5C=5A.

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Software and technical supports

1. Software download: please visit www.zketech.com to download latest EB Software.

2. Software function: refer to EB Tester Software User Manual packed in the software installation file.

3. Technical support: tech@zketech .com

4. Business consulting: sale@zketech.com

We will always adhere to the goals of "continuous progress and excellence", constantly upgrade the hardware or software, please often visit www.zketech.com for updates and instructions.

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